



Features of linear relationships

1 The relationship between two variables is linear when they change together _____. Tick **1** correct answer

☐ with a constant multiplier

☒ at a constant rate

☐ with the same difference

2 Does this table of values represent a linear relationship? Tick **1** correct answer

x	-2	-1	0	1	2
y	-2	3	8	13	18

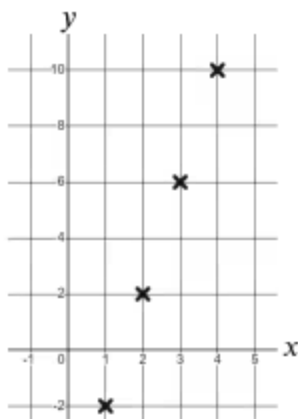
☐ No. The difference between -2 and 3 in y is different.

☐ No. x changes by +1 whereas y changes by +5.

☒ Yes. The y values change constantly by +5 as x changes by +1.

☐ We can't know until we plot it and draw a line.

3 What is the constant rate of change in this relationship? Tick **1** correct answer



☐ 1

☐ 2

☐ 4

☐ +1 in x and +2 in y

☒ +1 in x and +4 in y



4 Does this table of values represent a linear relationship? Tick **1** correct answer

x	0	2	4	6	8	10
y	6	3	0	-3	-6	-9

- ☐ No. The x values don't change by 1.
- ☐ No. The y values are decreasing, not increasing.
- ☒ Yes. The y values change by -3 for every change of $+2$ in x .
- ☐ We can't know until we plot it and draw a line.

5 Which of these rules will form a linear relationship? Tick **3** correct answers

- ☒ $y = 6x - 7$
- ☒ $y = 7 - 6x$
- ☐ $y^3 = 7 - x$
- ☐ $y = 6x^2 - 7$
- ☒ $6y = 7 - x$

6 Match the coordinate to the linear relationship it fits. Write the correct letter in each box

a	(3, 12)
b	(3, -12)
c	(6, -3)
d	(6, 10)
e	(8, 1)

d	$y = \frac{5x}{3}$
b	$y = 3 - 5x$
a	$y = 5x - 3$
e	$5y = x - 3$
c	$y = \frac{1}{3}x - 5$

